

YAGNA BHATT

+91 9662089636 | yagnabhattach00@gmail.com | Ahmedabad – 380008 | [linkedin.com/in/yagna-bhatt](https://www.linkedin.com/in/yagna-bhatt)

PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering student with hands-on experience in FPGA-based digital design, ESP32 wireless communication, Communication Protocols, Embedded C Programming and AVR microcontroller applications. Skilled in hardware-software integration and embedded systems. Active participant in e-Yantra Robotics Competition (eYRC) 2025, demonstrating strong technical problem-solving and teamwork in robotics.

EXPERIENCE

e-Yantra Robotics Program

2025

IIT Bombay

Remote

- Completed RTL design tasks using Intel Quartus, including Synchronous FIFO implementation and frequency scaling logic.
- Designed and verified FSM-based sequence detectors.
- Collaborated in team environments to analyze timing behavior and circuit performance for complex digital systems.

Hardware Design Intern

May 2026 – June 2026

Summer Internship

Ahmedabad, India

- Designed a modular chess clock system using Verilog HDL comprising Clock Divider, FSM Controller, Countdown Timers, and Pulse Debouncer.
- Implemented real-time timer control, player-turn management, and timeout detection through FSM-based architecture.
- Verified functionality using custom testbenches and waveform analysis in Xilinx ISim.

PROJECTS

FPGA-Based Switch Debouncing System

- Designed a Verilog-based module on Spartan-6 to eliminate mechanical bounce and ensure stable signal detection.

Wireless Error Detection & Correction System

- Built a robust communication system using ESP32 and ESP-NOW protocol to handle noisy transmissions.

Embedded Sensor Monitoring System using STM32

- Developed an embedded monitoring system using STM32 in Embedded C.
- Interfaced a temperature sensor and OLED display using I²C.
- Communicated sensor readings over UART and configured timers and interrupts for periodic sampling.

SKILLS

Programming Languages: C, Verilog, Embedded C

Hardware & Development: Spartan 6 FPGA, ESP32, ATmega16, Sensors & Actuators

Tools & Software: Intel Quartus, Proteus 8 Professional, Xilinx ISE, STM32 CubeMX, STM32 CubeIDE

Core Competencies: RTL Design, Wireless Communication, Signal Interfacing, Team Collaboration

Communication Protocols: UART, SPI, I2C, CAN

ACHIEVEMENTS

Verilog Programming Batch Winner

- Awarded for outstanding performance and top standing in the Verilog programming batch.

EDUCATION

Charotar University of Science and Technology (CHARUSAT)

Expected 2027

B.Tech in Electronics and Communication Engineering

Anand, Gujarat

- Completed 6th Semester CGPA-8.06

TECHNICAL EVENTS & PARTICIPATION

Techfest – IIT Bombay | *Pick and Drop Robotics Category*

- Participated in Asia's largest technical festival, competing in the robotics pick-and-drop challenge.

Techfest – NIT Surat | *Roborace Category*

- Competed in the Roborace robotics competition, demonstrating autonomous navigation and control skills.